

Team Details

Team name: **Aeroblast**

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Final Report

1. Introduction to Dataset and the Story

The dataset used in this project is the **Airfoil Self-Noise Dataset**, which contains aerodynamic and acoustic measurements of different airfoils.

The goal of this analysis is to understand how various aerodynamic factors such as **frequency**, **angle of attack**, **chord length**, **free-stream velocity**, and **suction side displacement thickness** affect the **Scaled Sound Pressure Level (SSPL)** — a measure of the noise generated by airfoils.

This analysis helps in identifying which design or operating conditions result in **lower noise production**, which is important in **aerospace and wind turbine blade design**.

Dataset Variables:

- **f**: Frequency (Hz)
- **alpha**: Angle of attack (degrees)
- **c**: Chord length (meters)
- **U_infinity**: Free-stream velocity (m/s)
- **delta**: Suction side displacement thickness (meters)
- **SSPL**: Scaled Sound Pressure Level (dB) — Target variable

2. Visual Analytics and Resulting Observations about the Story

(a) Histograms were plotted for all variables.

- Most variables show approximately normal distributions, but some are slightly right-skewed, especially **f** (frequency) and **SSPL**.
- The **SSPL** histogram shows a concentration of values between 100–120 dB, indicating most designs produce noise in this range.

(b) Scatter Plots (SSPL vs Inputs)

- **SSPL vs Frequency**: A positive correlation — as frequency increases, noise generally increases.

- SSPL vs Angle of Attack (α): Noise increases with larger angles, suggesting higher aerodynamic instability.
- SSPL vs Chord Length (c): Slight positive correlation — larger airfoils tend to be noisier.
- SSPL vs Free-stream Velocity (U_∞): Strong positive trend — higher airflow velocity increases noise.
- SSPL vs Delta: Moderate positive trend — thicker displacement layers lead to more turbulence and higher noise.

(c) Correlation Heatmap

- The heatmap shows U_∞ and f have the highest positive correlations with SSPL, implying that speed and frequency are key drivers of noise.
- The overall correlation pattern indicates aerodynamic parameters strongly influence the sound level.

3. Descriptive Statistics and Conclusions about the Story

The dataset contains **1503** observations across six variables:
 f , α , c , U_∞ , δ , and SSPL.

Variable	Mean	Std	Min	Max	Skewness	Observation	
f	2886.38	3152.57	200.0	20000.0	2.14	Highly right-skewed; wide range of frequency values.	
α	6.78	5.92	0.0	22.2	0.69	Slight right skew; moderate variation in angle of attack.	
c	0.14	0.09	0.025	0.3048	0.46	Fairly symmetric; small spread around mean.	
U_∞	50.86	15.57	31.7	71.3	0.23	Nearly symmetric; limited variation in flow velocity.	
δ	0.011	0.013	0.0004	0.0584	1.70	Strong positive skew; boundary layer thickness varies unevenly.	
SSPL	124.84	6.90	103.38	140.99	-0.42	Slight negative skew; sound pressure levels concentrated near the mean.	

Conclusions:

- Most input variables are positively skewed, meaning there are a few higher extreme values.
- f (frequency) and δ (boundary layer thickness) show the largest spread and skewness, suggesting they strongly influence noise levels.
- The target variable SSPL has a near-normal distribution (slightly left-skewed), suitable for linear regression modeling.

- Overall, the dataset is well-balanced with moderate variation, allowing for meaningful modeling and inference.

4. Inferential statistics with what inferences one can draw about the story

Hypothesis Testing (One-sample t-test)

Null Hypothesis (H_0): Mean SSPL = 110 dB

Alternative Hypothesis (H_1): Mean SSPL < 110 dB

Results:

- **T-test Statistic:** 83.37
- **p-value:** 0.0000

Although the p-value is very small, the observed mean (124.84 dB) is greater than 110, so there is **no evidence that mean SSPL is less than 110 dB**.

Linear Regression Modeling

A **multiple linear regression** model was fitted to predict SSPL based on the input features:

Model Performance:

- **R^2 Score:** 0.56
- **Root Mean Squared Error (RMSE):** 4.70

Regression Coefficients:

Feature	Coefficient
f	-0.001272
alpha	-0.405660
c	-34.469438
U_infinity	0.098021
delta	-139.460274

Interpretation:

- Frequency and displacement thickness have negative coefficients, implying higher frequency or thicker boundary layers may slightly reduce SSPL under certain conditions.
- Free-stream velocity has a positive coefficient, indicating higher airspeed increases noise.

- The R^2 score (0.56) suggests a moderate predictive strength.

Prediction and Quietest Design Analysis

The model was used to predict SSPL for new input settings:

Input Set	f (Hz)	α (°)	c (m)	U_∞ (m/s)	δ (m)	Predicted SSPL (dB)
1	800	0	0.3048	71.3	0.002663	127.63
2	1000	0	0.3048	71.3	0.002663	127.37

Quietest Design: Input Set 2 — *Predicted SSPL = 127.37 dB*

Overall Conclusions

- The dataset confirms that **airfoil noise** is influenced by aerodynamic and geometric factors, primarily frequency and flow velocity.
- Descriptive statistics establish that the average SSPL is around **125 dB** with low uncertainty.
- Inferential analysis indicates no statistical evidence of mean SSPL being below 110 dB.
- The regression model moderately explains variations in SSPL ($R^2 = 0.56$), showing potential for refinement using non-linear or polynomial models.

Code link: Url:

https://colab.research.google.com/drive/17RXYtzHx-DpKSQla3_p2BlqmlzsTd7tu?usp=drive_link